21-242 Recitation 12

Subspaces — Q1. Definition

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

Define "V is a subspace of W".

Course-level solution

Definition. A subset $V \subseteq W$ is a *subspace* of the vector space W if V is itself a vector space under the addition and scalar multiplication of W.

In practice one checks the subspace criteria: $\vec{0} \in V$, and $a\vec{u} + b\vec{v} \in V$ whenever $\vec{u}, \vec{v} \in V$ and $a, b \in \mathbb{F}$ (equivalently: closed under addition and under scalar multiplication).